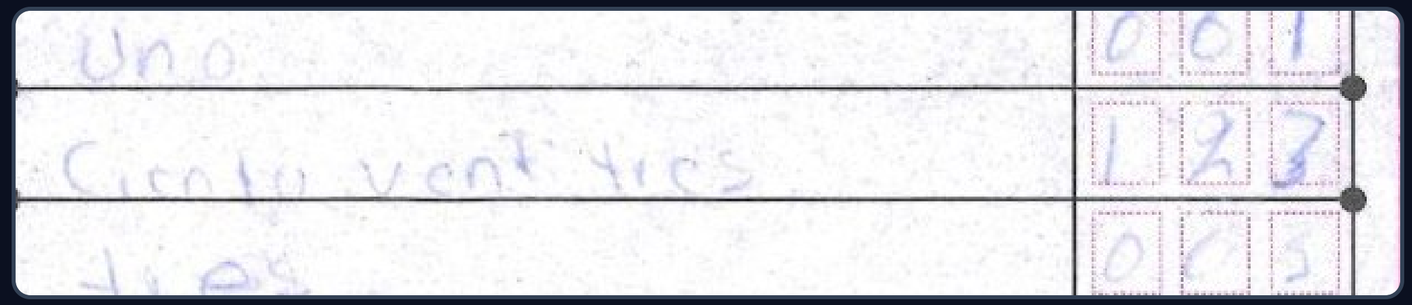
AMD DEVELOPER HACKATHON ACT III · GEMMA ON AMD TOKEN FACTORY

Two Hands

Verifiable quality control for handwritten numbers.

OCR reads the number. Two Hands checks whether the whole form agrees.



23
Gemma's read, digits and words

366 ≠ 466
rows sum vs the form's TOTAL

routed
to a person, with this crop

123
the paper; the sum holds

1,270
real handwritten tally sheets read on AMD

91.6%
field match vs **62.84%** for the reading the state bought

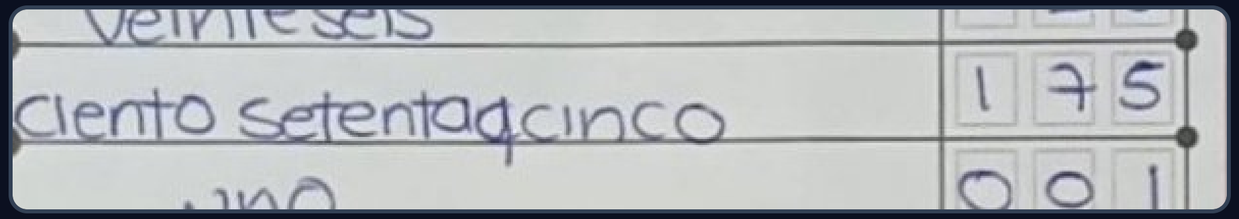
27 of 28
certified-detectable capture errors caught

$1
per sheet audited (proposed)

Form 122655C0100, Option 02: a real INE 2024 tally sheet, privacy-masked, cropped right of the logo column.

THE PROBLEM, FROM INE'S OWN FILES

Two official records of the same paper disagree. Neither says which one is right.



first capture keyed 75 **the paper says 175** **certified count: 175**

Form 050815B0100, Option 07 (masked, cropped right of the logo column). One of the 52 on our random 300.

- Every sheet is already keyed twice by people. Double keying copies the paper faithfully but **cannot say which paper is wrong**.
- The mandated audit checks **the software**, not the keyed values.
- A difference says two captures disagree. **The photo says which.**

6.5%

of the **52,167** polling stations not recounted: the preliminary capture and the certified count disagree on at least one vote field (**3,382**)

19.2%

of nationally captured tally forms fail their own row sum, in the official capture (**26,336** of **137,196**)

HOW IT WORKS

The model reads. The paper's own arithmetic decides.

1 · PHOTO

Scan or phone photo

Page located and **privacy-masked** (names and signatures blacked out) before it is stored or sent

→

2 · READ

Gemma on AMD

Gemma-4-31B-IT on AMD Token Factory reads every field **twice**: the digits and the amount in words

→

3 · CHECK

The paper's identities

words = digits · rows sum to TOTAL · section 5 = section 7 = TOTAL

→

4 · DECIDE

Accept or route

All hold: **accepted**, nobody keys it. One fails: **a person** gets the field named, with its photo

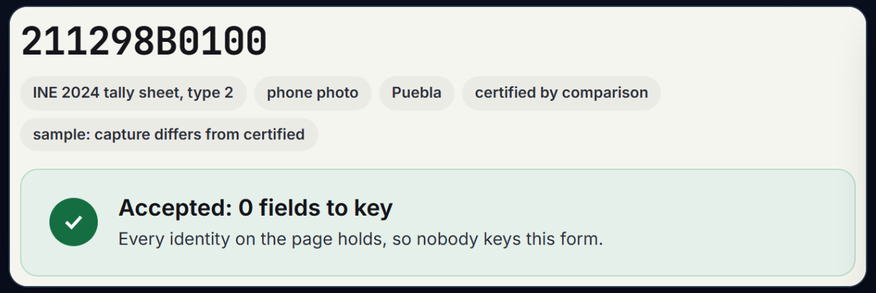
→

5 · RECEIPT

Sealed

Each field is a Merkle leaf; the report's root is anchored with **OpenTimestamps**

A second reader (Qwen3.8-27B, also on AMD) never overrides Gemma: it can only **withhold** acceptance



Form 211298B0100 on the site: every identity holds, so nobody keys this form.



Runs where

On AMD: every model read, incl. the second reader and the Evolus agent

On CPU: locating, masking, the gate

As: web app, API, CLI, MCP server

ONE REAL SHEET, REQUEST TO RECEIPT · /TRACE

Gemma read 23. The paper says 123. The sum caught it.

3. The gate

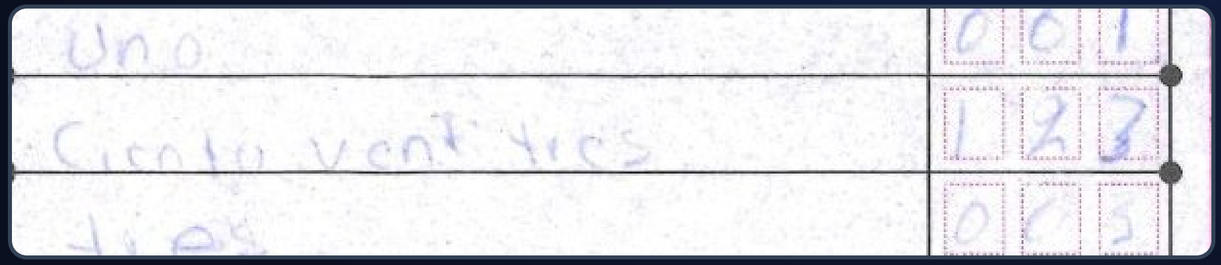
twohands/gate.py, run again on the raw output. Shape: 23 values, the form has 23. Row labels: consistent. Digits read: 23 of 23. Words: 23 of 23 fields with words read: digits = words (Option 02: digits = words).

IDENTITY	LEFT SIDE	RIGHT SIDE	RESULT
rows sum to TOTAL	**366** Option 01 1 + Option 02 23 + Option 03 3 + Option 04 26 + Option 05 22 + Option 06 98 + Option 07 165 + Option 08 0 + Option 09 0 + Option 10 0 + Option 11 0 + Option 12 7 + Option 13 0 + Option 14 0 + Option 15 0 + Unregistered candidates 0 + Null votes 21	**466** TOTAL 466	FAIL
votaron + representantes = section 5	**466** Voters on the list 466 + Representatives who voted 0	**466** Voters + representatives 466	pass
section 5 = section 7	**466** Voters + representatives 466	**466** Ballots taken from the boxes 466	pass
section 7 = TOTAL	**466** Ballots taken from the boxes 466	**466** TOTAL 466	pass

4. Decision

Routed to a person. Cause: rows sum to TOTAL: 366 != 466 (the only reason; the same as the stored record). Option 02: digits = words, so only the arithmetic could catch it. With the paper's 123, rows sum to TOTAL gives 466 = 466.

The gate, re-run on Gemma's stored raw output: one identity fails, so a person decides.



Gemma: 23 **366 ≠ 466** **routed** **paper: 123**

- **ok:** leaf recomputed from the record (`40c90012f177f089`)
- **ok:** crop SHA-256 matches the record (`0ddd5b7520df3724`)
- **ok:** path of 9 steps reaches the receipt's root (`e56a8ef4f011b9e5`)
- **ok:** receipt's root matches the root given (`e56a8ef4f011b9e5`)

VERIFIED. The receipt proves this record (form, field, keyed value, Gemma's digits and words, tier, crop hash) is unchanged since the report was sealed. It does not prove the reading is right: here Gemma's reading is wrong, and the receipt seals it all the same; the arithmetic is what caught it.

Leaf 146 of 267, Merkle root e56a8ef4f011b9e5. **VERIFIED** proves the record is unchanged since the report was sealed, not that the reading is right.

RESULTS · PRE-REGISTERED · INE'S PUBLIC DATA

1,270 real tally sheets. 49 capture errors confirmed by the certified count.

1,270
handwritten tally sheets read by Gemma on AMD

49
capture errors confirmed by the certified count, each with its photo

27 of 28
recall on the errors the certified count reveals (one state)

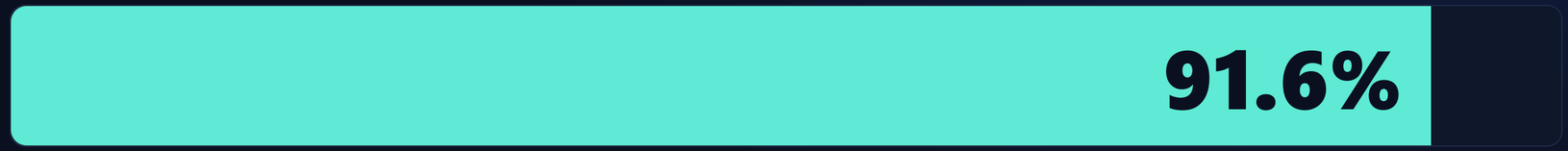
0 of 484
accepted fields wrong on 100 fresh forms (gate v2)

RUN	MEASURED
Pre-registered random 300	6,358 fields checked · 267 disputed · **52** where the official capture does not match the paper · 55 forms accepted with no person: 2 of 1,210 fields wrong
100 fresh forms (Amendment 3)	gate v2 accepted 22 forms: **0 of 484** fields wrong
Colima, a whole state	all 982 located presidential sheets for **$0.14** · 27 fields where the capture does not match the paper, each confirmed by the certified count
Recall (Colima, pre-registered)	**27 of 28** certified-detectable capture errors caught, 96% (Wilson 95% 82-99%); not total recall
A person's blind re-read	of the AI-judged disputes: **0 read differently**

AGAINST THE INCUMBENT · BAJA CALIFORNIA SUR, 2024 LOCAL ACTAS

91.6% field match, where the reading the state bought scored 62.84%

Two Hands: Gemma on AMD
pre-registered random 300 actas · 3,622 fields



Google Document AI
the state's purchased reading · all 2,020 actas



Same per-field exact-match metric. Ours: 91.58%, Wilson 95% 90.63-92.44%, against the published PREP values. Theirs: 14,501 of 23,077 fields, against the manual capture.

Reading is not the product

Ours accepts only what the paper's arithmetic proves and routes the rest to a person, with the photo.

Second keying by a person

The rule today. At least **0.8%** of fields in our random sample still do not match the paper (52 of 6,358).

Where a person looks

1 in 5 disputed fields is a confirmed capture error, against at least 1 in 122 at random.

GEMMA ON AMD TOKEN FACTORY

Why Gemma: more forms accepted, cheaper, faster

1.8×
the forms Qwen3.8-27B accepts, at the same error rate

4.6×
cheaper

2.8×
faster

4.8 /min
forms, 4 in flight, $0.000146 per form (shared endpoint)

SAME 300 FORMS, SAME GATE, ON AMD	FORMS ACCEPTED	WRONG OF ACCEPTED FIELDS	$ PER 1,000 FORMS	TIME PER READ
Gemma-4-31B-IT	**55 of 300** (18%)	2 of 1,210	**$0.14**	**87 s**
Qwen3.8-27B	31 of 300 (10%)	1 of 682 (+2 undecided)	$0.64	241 s
Gemma + Qwen3.8-27B, two readers must agree	89 of 300 (30%)	**0 of 1,958**	$0.78	–

Pairs: all 23 digits agree, lighter gate (secondary). Source: app/data/bakeoff_300.json. Every zero-error two-reader pair includes Gemma.

Evolus · workflow "Audit finding"

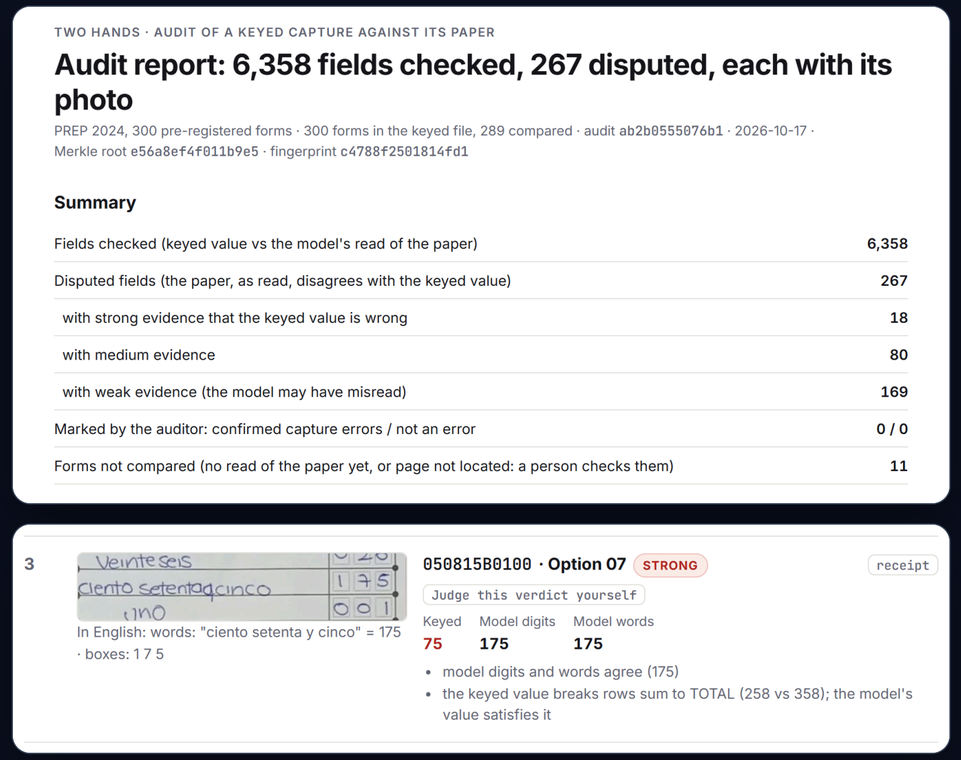
150 of 150 executions completed on real routed fields. The agent "Intake clerk" runs Gemma-4-31B-IT on AMD and drafts the finding; a Script recomputes the evidence score with no AI.

A whole state, then the country

Colima: all **982** located presidential sheets read for **$0.14**. About **$24** of AMD compute to read all 169,005 national presidential tally sheets.

BUSINESS AT A GLANCE

An independent check of the keyed values, $1 per sheet (proposed)



TWO HANDS · AUDIT OF A KEYED CAPTURE AGAINST ITS PAPER

Audit report: 6,358 fields checked, 267 disputed, each with its photo

PREP 2024, 300 pre-registered forms · 300 forms in the keyed file, 289 compared · audit `ab2b0555076b1` · 2026-10-17 · Merkle root `e56a8ef4f011b9e5` · fingerprint `c4788f2501814fd1`

Summary

Fields checked (keyed value vs the model's read of the paper)	6,358
Disputed fields (the paper, as read, disagrees with the keyed value)	267
with strong evidence that the keyed value is wrong	18
with medium evidence	80
with weak evidence (the model may have misread)	169
Marked by the auditor: confirmed capture errors / not an error	0 / 0
Forms not compared (no read of the paper yet, or page not located: a person checks them)	11

3

In English: words: "ciento setenta y cinco" = 175 · boxes: 1 7 5

`050815B0100` · **Option 07** STRONG

receipt

`Judge this verdict yourself`

Keyed	Model digits	Model words
75	175	175

- model digits and words agree (175)
- the keyed value breaks rows sum to TOTAL (258 vs 358); the model's value satisfies it

The product: the audit report. Every dispute with its photo, strongest first, sealed by one Merkle root.

BUYER

INE and the **32 state institutes**, each with an outside auditor (all 26 named in 2024 were universities)

WHY NOW

6 June 2027: the federal midterm and 32 concurrent local elections. INE budgeted MXN 9.5M for automated capture in 2024 and bought none

PRICE

$1 per sheet audited (proposed); first step: a **paid shadow-mode pilot** on one state's 2027 count. AMD compute: $0.00014 per sheet

REVIEWER WORK

About **one photo crop** per sheet to look at (0.92 measured), instead of re-keying about 22 fields

EXAMPLE

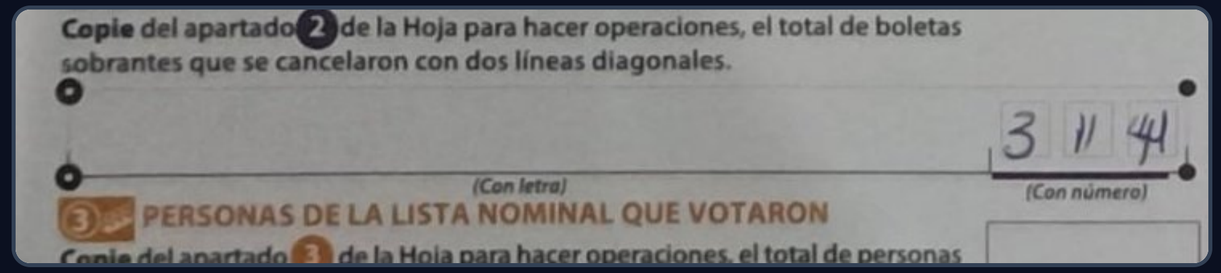
Morelos paid MXN 26.6M to run its 2024 PREP. Our check of every sheet: about **MXN 143k**, **0.5%** of that

NEXT

The same public data in **7 more countries**. Health reporting on **DHIS2** in 70+ countries. Not yet measured

WHERE IT HOLDS, AND WHERE IT BREAKS

Read 3114, paper 314: gate v2 sends it to a person



model read 3114 **paper: 314** **words line blank**

- Form 070212B0100, unused ballots: a field in **no identity**, with no words written. No sum can catch a digit inserted here.
- On 100 fresh forms (pre-registered), the frozen gate let it through. The v2 rule, **digits and words both read and equal**, blocked it and accepted **0 wrong of 484**.

Where it holds

One reader, frozen gate: **2 wrong of 1,210** accepted fields on 300 random forms (0.17%), **13 of 8,382** on a whole state (0.16%).

Where it breaks: the measured kinds so far

Harmonised in one call: 239/239 for 234

Blank words: 300 for 379

Digit insertion: 3114 for 314

Scope

Clerical capture quality only: whether a typed number matches its paper. No results, no parties (Option 01 to 15), no totals.

Most forms go to a person: one reader accepts 18%, two readers 30%.

The certified count is the official result. Not an official INE product.

CHECK IT YOURSELF

Two Hands at a glance: every number re-derives from public data

Reproduce

```
make eval
```

recomputes every number offline from the committed raw results and checks it against what the site shows

PROTOCOL.md: sample, prompt, gate and metrics committed before any model read; every later change is a numbered amendment

Verify a receipt

```
python -m twohands.verify_receipt
  receipt.json --crop crop.jpg
```

A changed value or crop fails.

Each report's Merkle root is anchored in Bitcoin with **OpenTimestamps**:

```
ots verify
  results/sample_audit_root.txt.ots
```

Proven on real paper

1,270 tally sheets read by Gemma on AMD

49 capture errors confirmed by the certified count

91.6% vs 62.84% for the incumbent reading

27 of 28 recall, one state, pre-registered

$1 per sheet audited (proposed)

Data: INE PREP 2024 and Cómputos Distritales 2024, free-use terms (Libre-Uso-MX) · masked pages only · an independent demonstration, not an official INE product

Two Hands. Verifiable quality control for handwritten numbers.